

MSYS 50 **ENTERPRISE ARCHITECTURE**

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Module 1: Enterprise Architecture

Module 1.1 – Overview of EA

DEFINITION

- Lankhorst
 - A **coherent whole** of principles, methods, and **models** that are used in the design and realisation of an **enterprise's organizational structure**, business processes, information systems and infrastructure
 - capturing requirements – from business and IT aspects
- Kotusev
 - collection of special documents artifacts describing various aspects of an organization from an integrated business and its perspective
Intended to bridge the communication gap between business and IT stakeholders, facilitate information systems planning and thereby improve business and IT alignment
- Bernard
 - analysis and documentation of an enterprise in its **current and future states** from an integrated strategy, business, and technology perspective
 - not just requirements in current form— CAPTURING requirements in its future state
 - “what does it want to be in the future?”
 - timelines to look out for: **short-term, medium-term, long-term**

Common standpoint

- Perspective – a lot of different
- Holistic – deals with entire organization
- Principles

- Practice
- IS/IT
- Business
- Integrated
- Documents – involves a lot of different to be produced referred to as “artifact”

Analogies

- Enterprise without EA = House without blueprints
- EA = City Planning
 - Evolving and dynamic

Additional characteristics

- Inherent complexity
- Need for structure
- Current and future state
- Continuously evolves
- Requires involvement of whole enterprise
 - Birds eye view
- Mutual understanding of diverse perspectives
 - Get message across, especially with clients
 - get everyone's perspective (developers, business)
- Communication is key

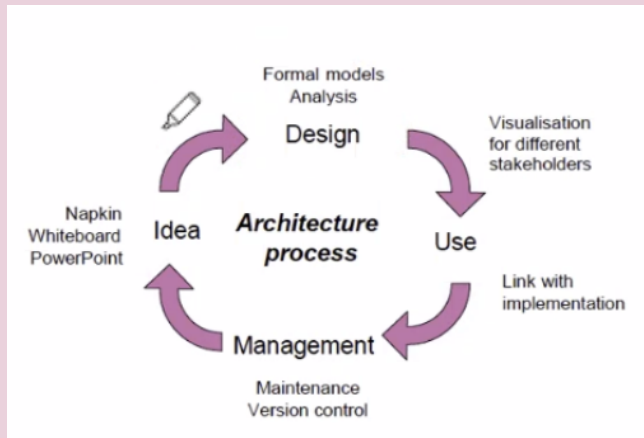
EA is NOT

- one-size –fits–all
 - each organization is different
 - Needs to achieve the business goals
- Purely technical planning
- About the artifacts
- Able to transform incompetent decisions/actions
- One-time planning
- Technology-specific
- About implementing frameworks
- Enterprise modeling/engineering

THE ARCHITECTURE PROCESS

- architecture is a process as well as a product

- Produced in a form of artifacts, models
- Idea → operational system → modify/replace
- More of a lifecycle, not linear
- Can start anywhere, iterative process



MAJOR ACTORS IN EA

- Business executives
 - higher ranks – overseeing all depts
- Business unit managers
 - dep managers, supervisor level
- IT executives
 - CTOS, IT/IS dep manager
- IT project teams
 -
- Architects
 - make sure to consider all perspectives
- Third-party (external)
 - Consultants, suppliers,

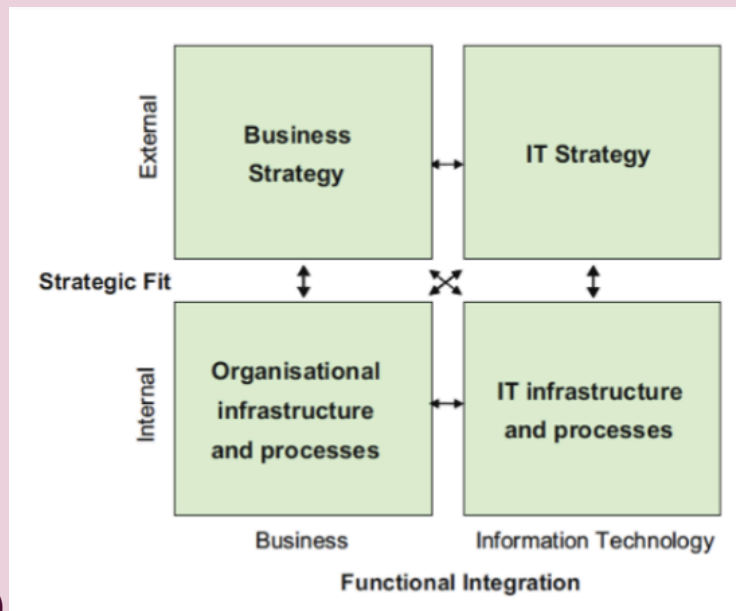
Module 1.2 – EA Drivers

Pressures that drive organizations

- Important to consider internal and external environments
- Good architectural practice indispensable, especially when organization grows and becomes
 - Try to create a roadmap that would help the org for their future growth

Internal Drivers

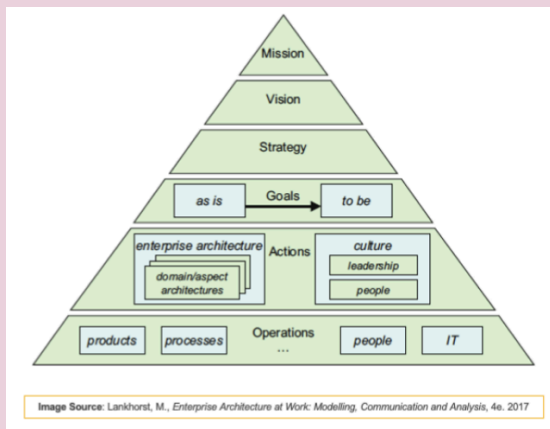
- Pressures Within
- Organizational alignment
 - Well-orchestrated interaction of organization components
 - Hard - well documented and formally disseminated; provides structure (job description etc...)
 - Soft - word of mouth & work environment; hard to measure (culture, friendship between employees, attitude)
 - Business-IT alignment especially crucial
 - How to align?
 - **Strategic Alignment Model (Henderson and Venkatraman**



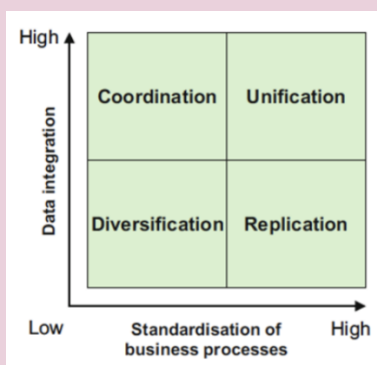
1993)

- Distinguishes between aspects of
 - Business strategy, organizational infrastructure
 - IT strategy, IT infrastructure
- Four dominant perspectives used for alignment
- Functional Integration = mixes both business side and IT side
- Strategic fit = needs to be able to support all sides
- Alignment Components (Nadler et al. 1992)
 - Work
 - People
 - Formal organization
 - Informal organization
- Organizational Alignment (Labovitz and Rosansky, 1997)

- Vertical (are they in touch with each other)
 - Top strategy
 - People at the bottom
- Horizontal
 - Internal processes
 - External customers
- EA as Management Instrument



- - mission: why
 - vision: where do we want to be – future vision
 - strategy: how to get there
 - goals: milestones to achieve
 - As-Is = current state → Goals/Actions → To-Be = desired future state
 - actions:
 - EA provides a roadmap that identifies the goals
 - Positioned within context of managing enterprise
 - Architecture is not a static asset
- EA Role (Ross et al. 2006)



- operating model – structure for how an org operates
 - An org probably belongs in one of the quadrants

- process standardization: how similar the process is
- data integ: data that is accessible throughout business units; how well it is used in outputs of each department
- Serves as organizing logic for business processes and IT infrastructure
 - Must reflect integration and standardization requirements

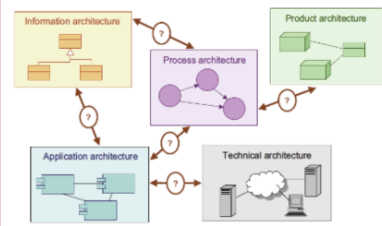
External drivers

- Outside the org
- May trigger push for adoption of an EA practice
- External regulations generate substantial impact
 - Disaster preparedness (like pandemic for schools)

Module 1.3 – the complexity of EA

- Understanding organizational complexity
 - Enterprise architecture is unlike building architecture
 - Orgs are dynamic socio-technical systems
 - Cannot be engineered then built
 - Complex, organic, living entities that gradually evolve over time
 - Successful orgs: well-managed evolutions ; well-organized
- EA and City Planning
 - Business executives – city governors
 - It project teams – construction teams
 - Architects – city planners
- EA Architects
 - Chief IT planners and owners of EA artifacts
 - Knowledgeable in business and IT, but have IT-centric background
 - Effective communicators
 - Team players
 - Innovators
- Responsibilities of EA Architects
 - Intermediaries between diverse stakeholders
 - Develop, update, peer-review, and approve EA artifacts
 - Create and maintain repository of EA artifacts

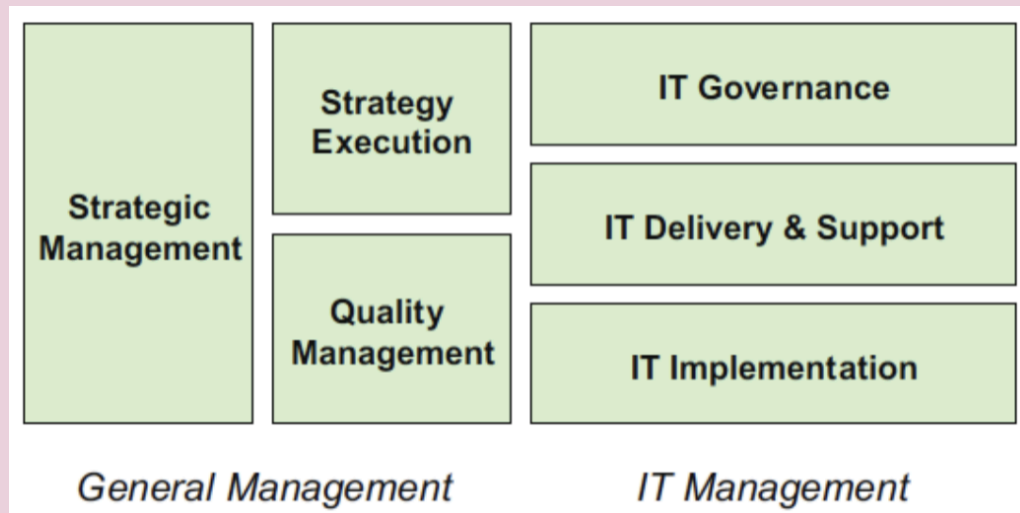
- Set-up software tools for EA
- Establish, run, and optimize EA processes
- Commonalities: Organizations and Cities
 - Cannot be perfectly planned in every detail
 - Future needs can be anticipated but not described in detail
 - Running and evolving, at the same time
 - Cannot be changed entirely, but through incremental modifications
 - Major changes do not happen overnight
 - Poor planning decisions can significantly hinder further development
 - Evolution is endless
 - No single best way to evolve
 - Planning must consider multiple concerns
- Major Considerations
 - Communication
 - Problem of alignment
 - Major challenge because each domain developed by distinct stakeholders with their own concerns and language
 - Needs bottom-up process
 - Integration of architectural domains
- Architectural Domains
 - Various organizational aspects important for business and IT (Kotusev)
 - Sub-architectures that describe distinct functional areas (Bernard)
 - Integration often a challenge
- Architectural Domains
 - depends per organization – have to talk to them to really see what their specific domains are

Kotusev ● Business ● Applications ● Data ● Integration ● Infrastructure ● Security	Bernard ● Strategic goals ● Workflows ● Dataflows ● Systems ● Infrastructure	Sample Architectural Domains (LANKHORST) 
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- Common Language
 - Use of common language is crucial
 - No need to start from scratch → better to use existing elements when possible
- Using Models as Representation
 - Models
 - Provide abstract, unambiguous definitions
 - Formal
 - Many types
 - Help capture the essence of EA
 - Only core of each domain should be included
 - Multiple models usually needed
- Architectural Frameworks
 - Integrates different elements
 - Many available for EA
 - Considerations
 - Is it applicable and/or useful to the organization?
 - Is it worth adopting one? If not, what is the alternative?
 - template / guide
 - there are other approaches – non-framework approach
 - Kotusev

Module 2: The Business-IT Dynamics

Module 2.1 – EA and Other Governance Instruments



Relevant Business- and IT-Related Standards

- Balanced Scorecard
- Business Model Canvas
- Business Architecture
- European Foundation for Quality Management
- ISO 9001

Strategic Management

- Kaplan and Norton (1992): financial measures are inadequate
- Strategic Planning:
 - M/V statement
 - goals
 - swot analysis is also used
- Balanced Scorecard (BSC) clarifies vision & strategy (**4 perspectives**):
 - Customer
 - Financial
 - Internal Business Processes
 - Learning and Growth

associated with:

- Three-layered structure for each perspective (are they still applicable?)
 - Mission

- Objectives – QUANTITATIVE
- Measures – QUANTITATIVE
- Quantitative perspectives- most impactful to EA
- EA especially useful in last two perspectives
- Example: Balanced Scorecard for an Academic Institution (**BSC**):

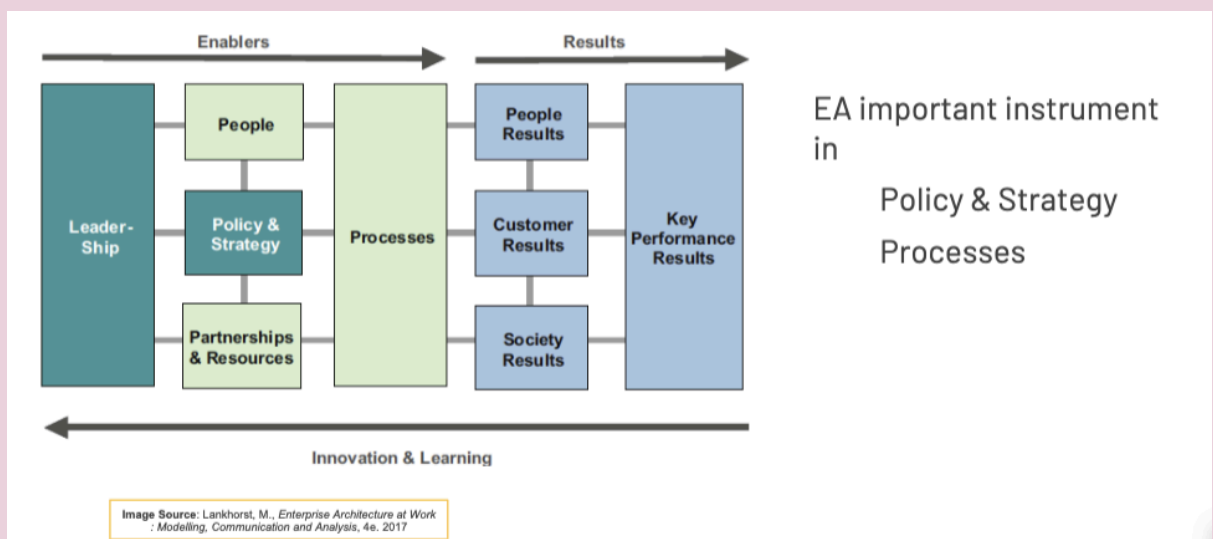
Perspective/ Layers	Mission	Objectives	Measures (e.g., KPIs)
Financial	To steward resources effectively to support accessible education.	Ensure long-term financial sustainability and resource allocation.	Tuition revenue per student, fundraising effectiveness, cost per degree awarded.
Customer (Student)	To empower students to achieve their full potential and serve the community.	Enhance student success, satisfaction, and community engagement.	Student retention rate, graduation rate, post-graduation employment outcomes.
Internal Business Processes	To deliver excellence in teaching, learning, and research.	Improve the efficiency and quality of core academic and administrative functions.	Curriculum design quality, teaching effectiveness, administrative efficiency.
Learning and Growth	To foster an inclusive, innovative environment for continuous learning.	Develop faculty, staff, and technology to support institutional improvement.	Faculty development participation, staff satisfaction, technology integration.

Business Model Development

- Business Model Canvas (Osterwalder and Pigneur 2010)
 - Provides template to create high-level description of new or existing business models
- 7 parts
- Intuitive layout
- Often used during brainstorming sessions
- Many canvas elements can be translated to EA models

European Foundation for **Quality Management**

- European Foundation for Quality Management (EFQM) is an important framework
 - Broader scope than ISO 9001
 - Benchmarks organizational performance with best practices
 - Complements BSC
 - Provides principles, measures, and indicators for assessing performance of organization → becomes basis of continuous learning, innovation, improvement
 -
- Example: **EFQM** Criteria for Excellence



- handles policy making and how processes are implemented → actions and operations — THESE DIRECTLY AFFECT ENTERPRISE ARCHI
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ISO 9001:2000

- Outlines criteria for a good quality management system (**QMS**)
- QMS designed and documented to control how processes are performed
- Product realization: business processes that realize company's product/service
- Key processes: those that affect product/service quality
- Once QMS is installed, audit may be requested for certification
- EA helps conform to ISO 9001 on process identification and documentation

IT Governance

- **COBIT** (Control Objectives for Information and Related Technology)
 - admin
 - predictability
 - info flow → guidance
 - restrictions
 - maintenance
 - rules & regulations
 - standards
 - **SECURITY**
- this complements directly to EA – since its in actions/operations level
- IT control framework that aims to bridge gaps in business risks, control needs, and technical issues
- Also provides
 - Critical success factors for achieving optimal control over IT processes and key goal indicators
 - Maturity model for IT governance, similar to CMMI
- EA a natural complement to COBIT
 - EA focuses on business & IT structures and processes, COBIT on how IT should be organized

IT Delivery and Support

- IT Infrastructure Library (ITIL) widely accepted best practices in IT service delivery
- Comprised of documents that give guidance on provision of good IT services
- Broad groups of processes
 - Service delivery
 - Service support
- Complementary: COBIT's control objectives tell what to do, ITIL explains how
- EA provides clear understanding of IT, infrastructure, business processes → crucial to ITIL's focus on management of IT assets

IT Implementation

- **Capability Maturity Model Integration (CMMI)** enables assessment of maturity of software engineering processes; also provides key practices to help increase maturity
 - ASSESSMENT TOOL
 - ASSESS MATURITY OF SOFTWARE ENGINEERING
- In 2000, upgraded from SW-CMM with expanded scope

- Five maturity levels
 1. Ad Hoc – no structure in place
 - 2.
 3. Defined –
 - 4.
 5. Optimized – everything is in place + optimized
-
- EA provides context, constraints, and guidelines for individual software projects
 - EA especially useful from Level 3 and beyond
- Assessment tool

Module 2.2 – The Problem of Business and IT Alignment

Strategic Management Outputs

- Strategic management process produces high-level business strategy defining long-term direction
- Business strategy
 - Mission
 - Vision and values
 - Strategic goals and objectives
 - Competitive & environmental analysis (e.g. SWOT)
 - Key Performance Indicators (KPIs)
- *“Does it make sense to start information systems planning directly from the overall business strategy to achieve alignment?”*
 - Business Strategy is Inadequate
 - Vague, unknown, or merely absent
 - Rarely provides clear direction for IT
 - Unstable and frequently changes
 - Often requires non-reusable IT components

Business Strategy Problem #1

- Generic statements
 - To be the leading ISP in the country
 - To increase market share by 15% over the next three years
- Subject to different interpretations

- May not be entirely documented
- How are these useful to IT planning?

Business Strategy Problem #2

- Business strategy does not often contain specific suggestions regarding practical IT implementation
- Business-related targets and performance goals offer little guidance
- IT planners often forced to decide based on guesswork
- Little to no value as input to IT and EA

Business Strategy Problem #3

- Frequent changes in business strategy gives no solid basis for IT planning
- Priorities could shift at any time
- Wreaks havoc on existing IT plans

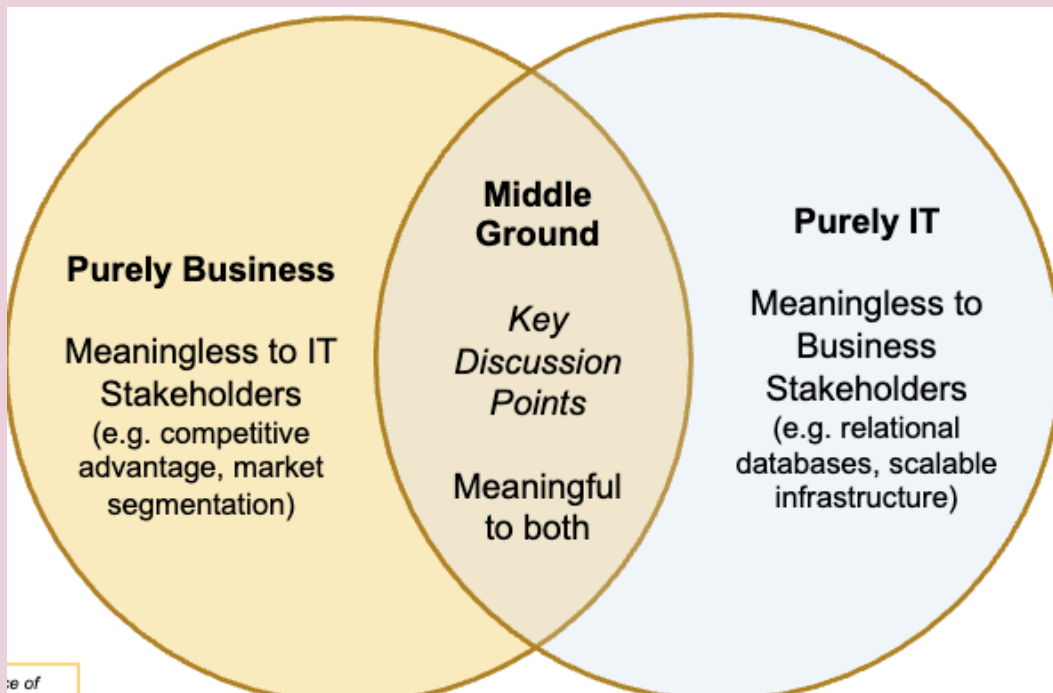
Business Strategy Problem #4

- Many IT systems only beneficial in short term
- Systems lifespan complicates modifications and IT agility
 - Current systems are assets, but may be liabilities in next strategy
- Isolated legacy systems could accumulate
- Focus on new systems for current strategy could lead to ***alignment trap***
 - Alignment Trap:
 - hiring ppl n creating deps specific only for a certain goal / system
 - when it gets scrapped, they won't get used anymore – rendering them useless

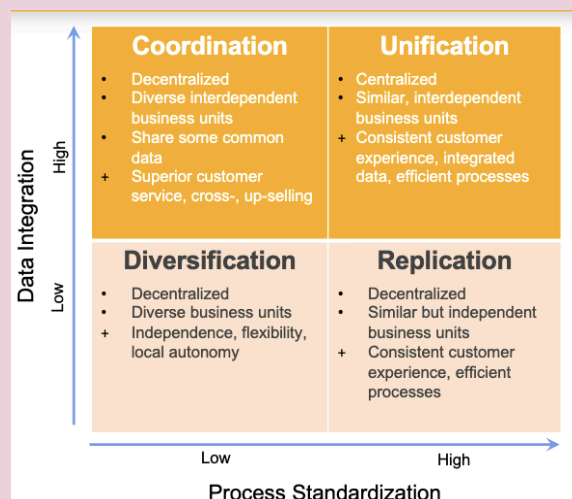
What Needs to Happen

- Middle ground between business and IT
 - Addresses concerns on both sides
 - Understandable and valuable for both
- Business: explain value of IT decisions
- IT: provide actionable suggestions
- Should be valuable either in short- (<1-2 years) long- (3-5 years) and very long-term (>3-5 years)

Key Discussion Points Between Business and IT



- e of
- Operating model
 - Desired organization-wide levels of
 - Process standardization
 - Data integration
 - Discussed in Ross et al. 2006 paper, as mentioned previously
 - Implies standardization of key business-enabling EA domains
 - Creates four major models based on combination of desired levels

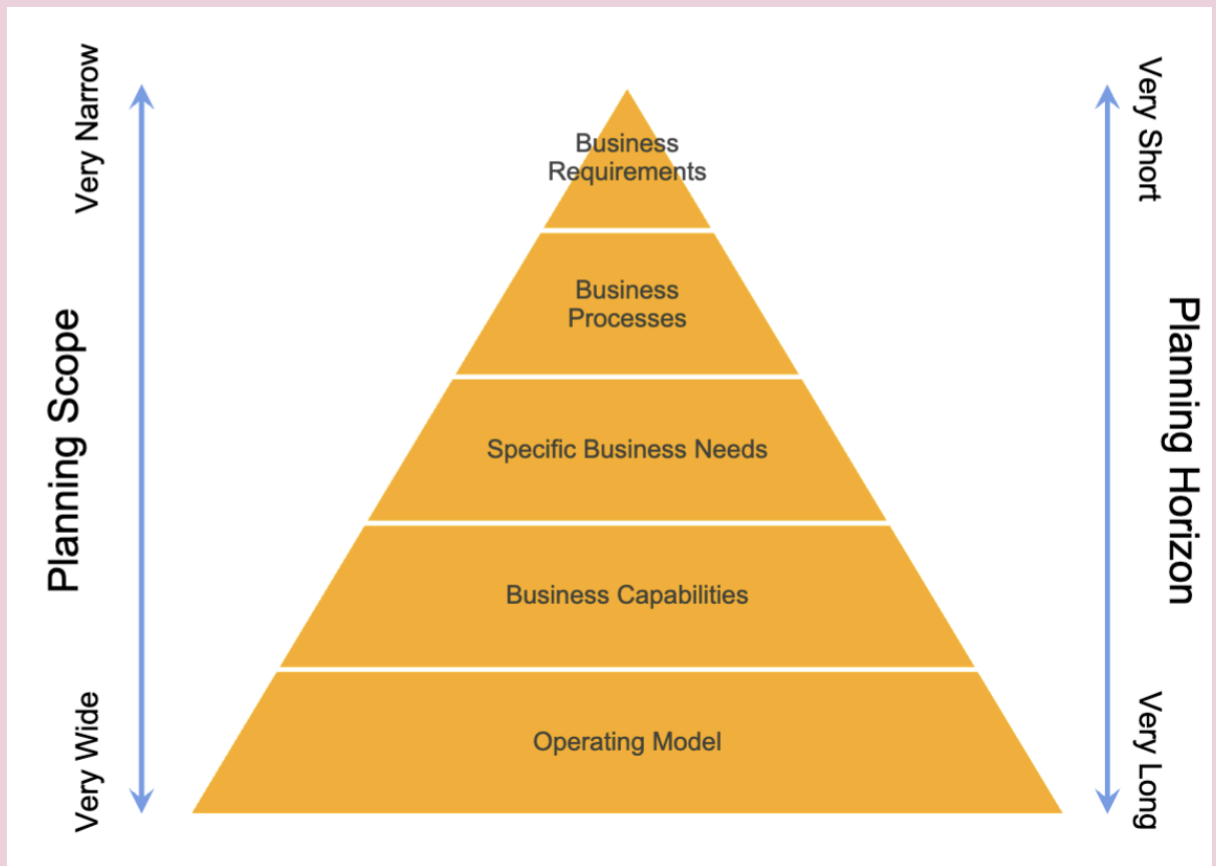


- - diversification:
 - replication: resto
 - Diversification: Acquiring businesses unrelated to core business

- Replication: Same business processes when moving from one department to another. High on process standardization. Thus, ensures consistency across multiple business units. (Ex: Hotel Chains)
- Coordination: Equal understanding of data but independent business units
- Unification: Centralized unit
- Ex: Disney (Coordination)
 - Low process standardization because of multiple companies (hotels, theme parks, animation film company)
 - High Data Integration because of their branding and are less autonomous in terms of customer experience.
- Ex: Airline Businesses (Unification)
 - Experience and data needs to be consistent
- Is it possible for businesses to switch to another quadrant in the operating model?
 - Yes.
 - Ex: Amazon
 - Originally focused on selling books = Unification
- Shapes overall reporting structure and defines level of autonomy
- Organizations need to decide on an operating model to thrive in its environment
- Choice of model reflects fundamental considerations not expected to change in foreseeable future
- Considered most abstract of all discussion points
- IT planning horizon: very long-term
- Business capabilities
 - General ability of organization to perform business activity
 - Example: customer behavior tracking
 - Encompasses underlying business processes, people, knowledge, and other resources
 - Considered consistent and form building blocks to organization's business functions
 - Major changes due to shift in core business model
 - Business executives clear with strategy-capability relationship

- Planning horizon: long-term
- Specific business needs
 - Concept of addressing specific business problem using IT to gain improvements
 - Example: improve accuracy of tracking employee work hours
 - Convenient discussion point for business and IT
 - Planning horizon: medium-term (2-3 years)
- Business processes
 - Sequence of activities carried out by specific actors, with specific inputs and outputs
 - Enables identification of what IT systems might need to be deployed
 - Approximates how solutions should work and which activities need to be changed
 - Planning horizon: short term
- Business requirements
 - Detailed functional and non-functional requirements for a specific information system
 - Reflect desired behavior of a system
 - Most detailed among discussion points,
 - Planning horizon: actionable within 6-12 months

- Pyramid of Key Discussion Points



Module 2.3 – The IT Initiative

What are IT Initiatives?

- Key unit of work in EA practice resulting in a deployed IT solution or desired modification in IT landscape, consequently enabling business improvements
- Typical purpose: address a specific business need
- Initiative may be
 - Single project
 - Multi-project (w/ unified purpose and business case)

Attributes of an IT Initiative

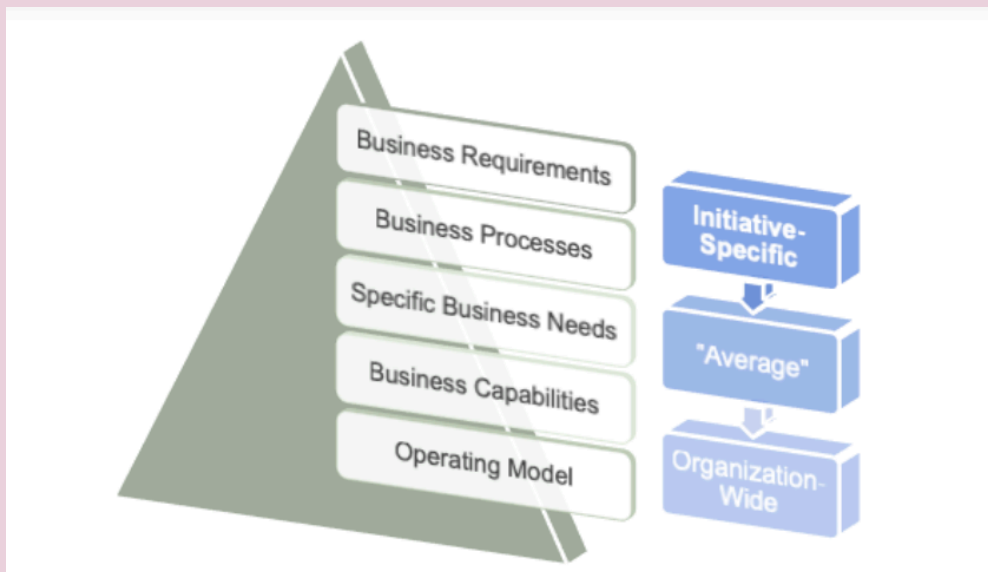
- Core business need or problem
- Main goals and objectives
- Scope
 - What stakeholders will use the system?
- Business case (financial details including initial investment, **ROI**)
- Business sponsors

- Budget

Context of IT Initiatives in EA Practice

- Understand **what** IT initiatives should be implemented in the future
 - Optimal initiatives portfolio for maximum business value
 - Identification of: Current & future initiatives
 - prioritization
 - which should be proposed first
 - which should be deployed first
- Understand **how** these initiatives should be implemented
 - Optimal ways to deliver
 - Implementation aspect
 - Options: In house Development; outsourcing; off-the-shelf,

Discussion Points and Level of Detail



- Operating Model determines business capabilities
 - Specific Business Needs
 - IT initiatives

Five Types of IT Initiatives



- Projects, possible improvements

1. Fundamental Initiatives

- Addresses fundamental or “permanent” business needs
- Strategy-agnostic
- Always necessary and useful for the business
- Identified top-down by C-level executives and architects, often from requirements of operating model

regardless of what models the org has, fundamental initiatives stay the same

only changes if it changes industry

2. Strategic Initiatives

- Stems from **current business strategy**
business strategies could shift – which means it has no solid foundation
BUT – using Risk Assessment Management:
(using probabilities)
 - if it works, it gives a competitive advantage to the company
 - *high risk, high reward.*
 -
- May be proposed directly by top executives for immediate impact on strategy execution
- Caveat: danger of becoming legacy after current strategy

3. Local Initiatives

- a. Addresses specific local business needs
- b. May be unrelated to organization-wide strategy
- c. Proposed bottom-up by local executives
- d. Reviewed by top executives and architects based on importance to whole organization
- e. Approved local initiatives prioritized

4. Urgent Initiatives

- a. From urgent business needs (e.g. overlooked critical problems, changes in government regulations, new operational demands, etc.)
- b. Unanticipated, unpredictable, unplanned
- c. Usually triggered by external business environment
- d. Bottom-up
- e. Reality: substantial in number

5. Architectural Initiatives

- a. Aims to improve overall quality of organization's IT landscape
- b. Does not directly address specific business needs (unlike other types) or provide new business functionality
- c. Top-down by architects
- d. Examples
 - i. Upgrade obsolete hardware
 - ii. Decommission legacy systems

Module 3

Module 3.1 – Architectural Complexity and Domains

Support Needed at Various Phases of Architecture Life Cycle

- Capabilities needed to have a high level success
- **Design**
 - Use of common, well-defined vocabulary
 - Does not focus on single domain, but allow integration of architectures from different domains
 - Build upon Natural everyday language used to communicate with colleagues with technical terms known to organization or key domains
 - Compilation of organization-wide guidelines, best practices, and other standards
 - Support for tracking architectural changes and decisions
- **Communication**
 - Architecture shared with various stakeholders from within and outside the organization
 - Need to visualize relevant aspects for a particular group of stakeholders
 - Very important to emphasize relations (connections and dependencies) among different domains
 - Need to be able to ***Filter*** Information
 - stakeholders usually care about their own concerns
 - Determine connections and interactions between domains
- **Realization**
 - For better implementation/realization of design, design considerations must go on a more detailed level
 - Seamless transitioning between levels
 - Integration between different design tools should be provided
- **Change**
 - Change in one architecture may affect others in different domains
 - Assess consequences of such a change
 - Can snowball

Compositionality

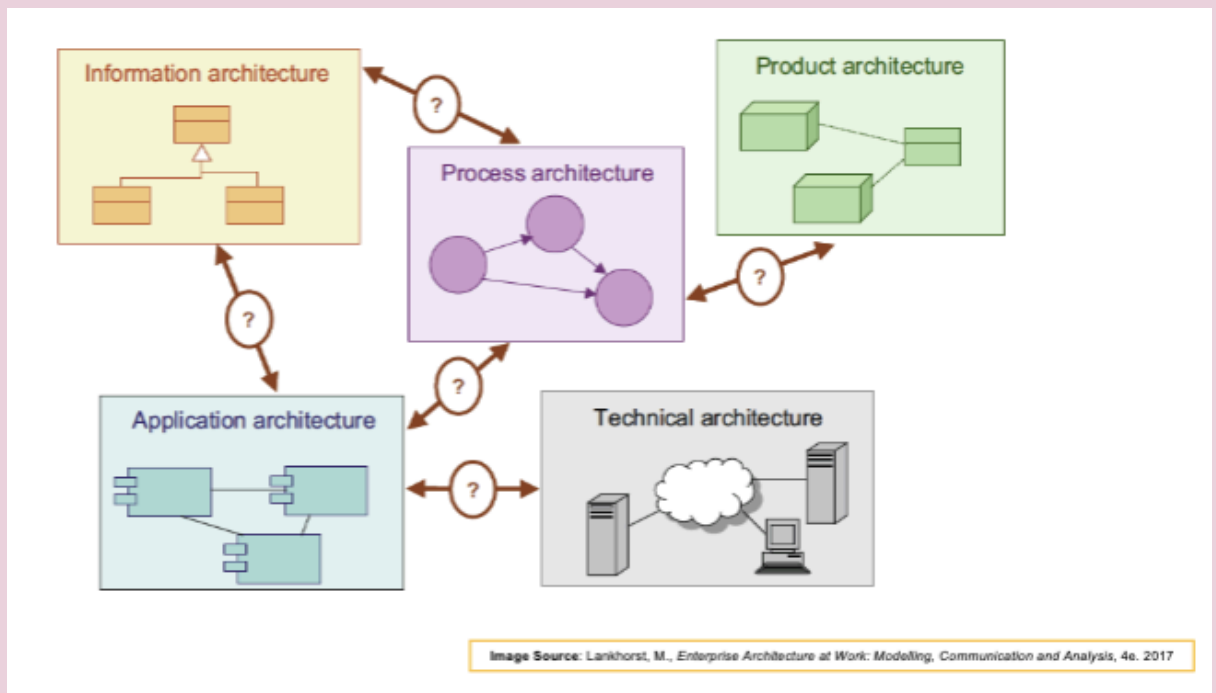
- Standard way of dealing with complexity is via a compositional approach
- Distinguishes different parts of a system as well as relations between parts
- Example
 - Understand different parts of a car and how they work together as a whole
 - Breaking down things first into more manageable parts (architectural domains)

Integration of Architectural Domains

- One way to deal with EA complexity
- Provides insight for those who deal with architectures
- Some integration problems cannot be solved
- Cannot eliminate integration problems
 - Concepts and tools only make it easier to deal with issue
- Two typical integration problems shown next

Integration Problem Example #1

- Different modeling standards and languages (e.g. UML, BPMN) used for each architectural domain
- View of each domain developed by distinct stakeholders with their own concerns
- Challenge to do one-to-one mapping (alignment) of model elements from different domains
- Example:
 - Different notations, even within IT: entities, relations vs. classes, objects



Integration Problem #2

- Integration of structural and behavioral models
 - Structural models: fundamental components of the system
 - Behavioral models: reaction to those components but can affect structural models as well
 - Both models affect each other
 - Ex: Changing the water pressure = affects the structural integrity of the faucet (it may break), or the structural integrity of the faucet may affect the output pressure of the water
- Example
 - How are software components related to application functions?
 - ERD and DFD: how are they related?

Choosing Architectural Domains

- Domain: area of knowledge, influence, or ownership
- Framework-dependent domains
- Traditional vs. Modern (i.e. Cloud)
- Could start with Business and IT → possibly break down further

Questions to Ask When Choosing Deciding on Domains

- Who are the major stakeholders? What would each one want to focus on?
- What are the priority areas of the organization?

- What are the capabilities of the organization? What are the strengths and weaknesses of the organization?
- Is the preferred architectural approach flexible enough to freely allow choice of domains?
- How easy is it to integrate the different chosen domains?

Module 3.2 – Describing Enterprise Architecture

Capturing the Essence

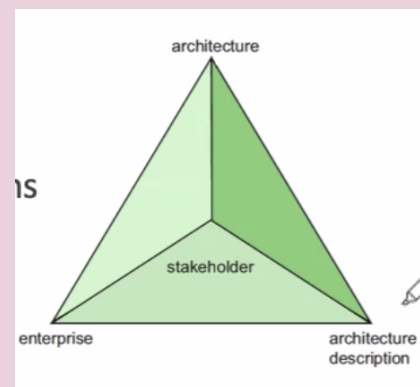
- Essence of each domain is of more value to EA
- Focus on relations and interactions between domains
- Use of models is indispensable

Model

- abstract and unambiguous conception of something in the real world
 - minimize miscommunication
 - *you're sending a message to the stakeholders – don't want to be sending an ambiguous message*
- uses **formalized graphical** or textual **language**
 - keep the number of symbols low – don't want to overwhelm stakeholders. they want to understand the models.
- Model's rigor requires architects to be more meticulous
- difficult to capture everyone's needs in single model

Observing the Universe

- Enterprise compared to universe
 - establishes the context of the organization
- Viewer (stakeholder) articulates conception of universe
- Conception is architecture
 - stakeholder group's understanding of their part of the organization
 - **Architectural Conception:**
collection of intangible (undocumented) understanding based on the perspective of each stakeholders



- capture this in form of models / artifacts
- ex.
 - AS STUDENTS OF ADMU, What is your understanding of ADMU?
What are your ideas of ADMU? – Understanding of ADMU
 - not documented- but a conception of what ADMU is
- Representation
 - **Architecture description:**
capture this collection of ideas in form of models / artifacts
collection of intangible (undocumented) understanding
based on the perspective of each stakeholders
 - tangible representation of the architecture itself in artifacts
 - models, textual descriptions

Concerns

- Stakeholder's interests influence conception of a part of the enterprise
- Stakeholders may be regarded at individual or aggregated level (e.g. business manager as stakeholder, also entire business management as another)
- Stakeholder's preconceptions (social, cultural, educational, professional background) also a factor
- Therefore, assume models are created from certain perspective
- ❖ could be diff perspectives

Observing Domains

- Domain
 - Subset of conception of universe
- Model
 - Abstracted and unambiguous conception of a domain
- Modeling
 - Act of abstracting a model from a part of universe
 - Uses some language and medium

Views and Viewpoints

- Most stakeholders not interested in full scope of architecture, except architects
- Stakeholders require specific views
- View
 - Specified through a viewpoint

- Visible elements in a viewpoint dependent on stakeholder
- View is what one sees, viewpoint tells where one is looking from
- **Viewpoint:** A PERSPECTIVE.
 - where one is looking from
 - not complete perspective
 - ONLY FROM A CERTAIN PERSPECTIVE – usually stakeholder
- **View:** VISIBLE ELEMENTS.
 - all elements that they can see based on their perspective
 - *what can you see from your perspective?*
 - collect views from different perspectives, diff stakeholders, diff domains
- Other domains will tell the whole story

Architecture Models

- Related ways of thinking, modeling, communicating, working, supporting, and using
- Single model for enterprise is ideal but infeasible in reality
- In practice, partial models from different domains make a collective whole

Key Points about Models

- Pictures and diagrams should not be equated with models
 - pics: can be subjected to *multiple interpretations*
 - models: *only one interpretation*
- Important to distinguish between
 - **Content** – concepts involved
 - visible elements to put into model
 - allows architect to specify the scope of the model
 - **Visualization** – how content is presented
 - representation of the content
 - *easiest and efficient way for stakeholders to understand and appreciate the content*
 - make it appealing to the stakeholders
- Architecture should be independent of visualization
- Architecture semantics
 - Inherent meaning
 - Two distinct views (of the same thing) have common elements

- Consistency is advantageous
 - consistent across models
 - models that cut across models !!
 - problems arise when it comes to transitioning models into physical

Semantic and Symbolic Models

- Symbolic model
 - Represents architecture properties using symbols that refer to reality
 - Uses formal syntactic structure and notation known as description language
 - **PROBLEM:** *can be technical for other groups of stakeholders to understand.*
- Semantic model
 - Represents one possible interpretation of a symbolic model
 - **MORE IMPORTANT IN EA:** *easier for stakeholders to understand since symbolic models can become too technical.*
- UML focuses on symbolic model; ArchiMate emphasizes latter

Module 4

Module 4.1 – Architecture Approaches, Methods, & Frameworks

Approaches

- Core Strategies
 - Frameworks
 - Non-frameworks based
 - Hybrid
- Relations between architectural domains must remain clear
- Changes should be tracked

Framework-Driven Approaches

- helps structure architecture description techniques by identifying and relating
 - architectural viewpoints, and

- associated modeling techniques
 - TOGA
- Framework may provide process, key deliverables, metrics, etc

Framework use not a guarantee

- Lankhorst
 - Adoption of a certain framework is not sufficient
- Kotusev
 - “Popular EA frameworks are useless products or marketing specialists and management gurus”
- *mere use of frameworks are not enough*
- *some frameworks – given the proper context w/ the business needs context, etc – is very useful*

Popular Frameworks

- The Zachman Framework
 - John Zachman; 1987
 - Originally: framework for Info System Archi
 - Logical structure for classifying and organizing descriptive representations of an enterprise
 - *classification structure only; organizes artifacts*
 - *no tech, no milestones*
 - *basically like an excel sheet*
 - Design artifacts at intersection between
 - roles
 - owner, designer, builder
 - product abstractions
 - what, how, where

	what	how	where	who	when	why
planner						Scope (conceptual)
owner						Enterprise model

						conceptual
designer						
builder						
sub contract or						

- for completeness, additional abstractions may be used
 - WHO WHEN WHY
- Advantages
 - easy to understand
 - addresses enterprise as a whole
 - covers the whole enterprise
 - independent of tools/methodologies
 - helps to identify issues
 - issues can be mapped to understand where they fit
- Disadvantages
 - large number of cells
 - relations between cells not well specified
 - altho this one - di naman. pretty clear.
- Using the zachman framework
 - Identify focus areas, concerns, priority
 - Dont do all agad to not get overwhelmed
 - Top down or bottom up
 - Aggregate relevant existing artifacts, identify those that must be developed
 - Start with foundation: minimum viable architecture, starting with cells critical for decision-making
 - minimum viable architecture, composed with all the artifacts
 - Possibly pair with another approach
 - adopt the methodology of TOGAF
 - mixing and matching of frameworks
 - possible to only use the zachman framework

- *framework can be adjusted to only what's needed for the enterprise*
 - *can be 10 cells only*
- TOGAF
 - The open group architecture framework
 - Originated as generic framework and methodology for development of technical architectures
 - treats the enterprise as a system
 - from version 8 onwards
- Model-Driven Architecture
 - By the object management group (OMG, 2014)
 - Aims to provide open vendor-neutral approach to interoperability
 - *it doesn't care about who manufactured the hardware*
 - Raises level of abstraction for specifying software solutions
 - IT-centric (software development and deployment) to be considered an EA Framework
 - *very useful for automating systems within enterprise' IS.*
 - *nothing much to do with business*
 - *focus is really only IT*
 - *developing and deploying for the IT*
 - three levels of abstraction
 - LVL 1: computation independent
 - independent of Automation or Manual
 - *most basic level*
 - LVL 2: platform independent
 - agnostic of platform – agnostic of hardware/software
 - you're going to start automating , just doesn't care about the platform used
 - LVL 3: platform specific model
 - what platform – hardware, software, operating systems will be used
 - What will be automated?
 - *most detailed level*
 - notion of mapping is a key feature
 - set of rules and techniques to translate one model to another